

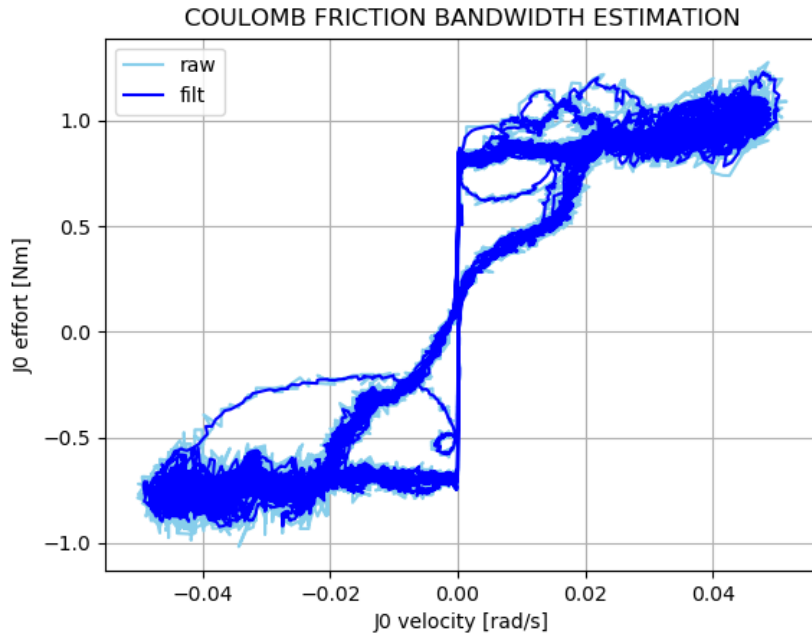
Joint 0

Python frequency = 250 Hz

ROS frequency = 1 Khz

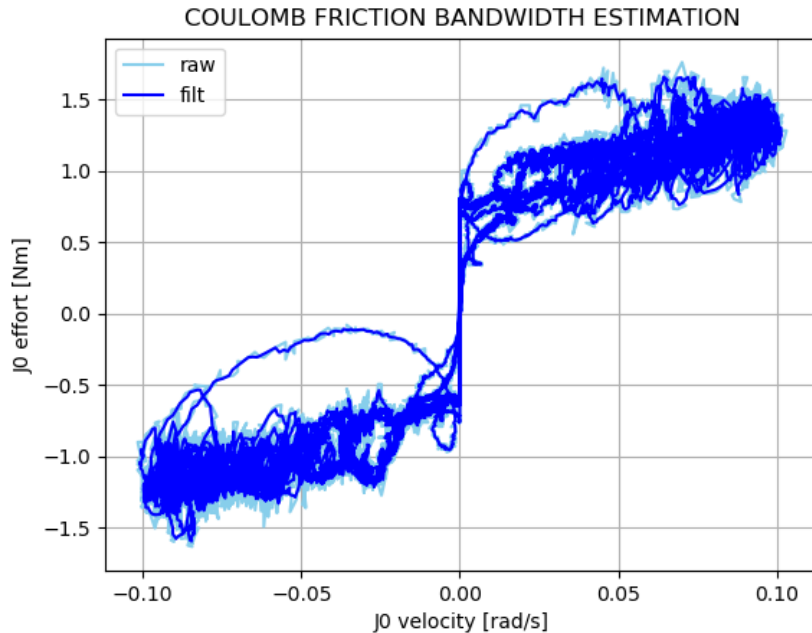
servo_jp

Joint 0



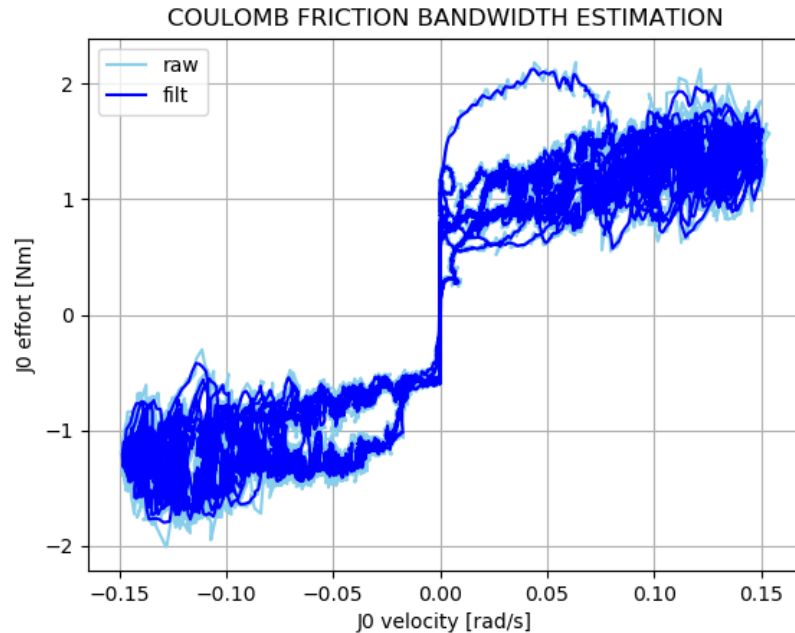
- acc = 0.1 rad/s
- seg_duration = 1s
- center = 0 deg
- amplitude = ± 0.7 deg
- J1 = 0, J2 = 0.12 m

Joint 0



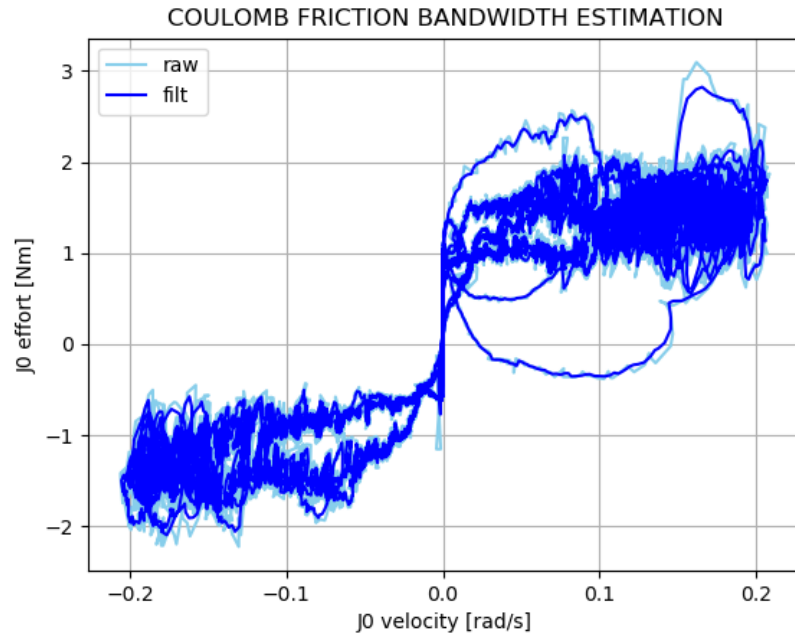
- acc = 0.1 rad/s
- seg_duration = 2s
- center = 0 deg
- amplitude = ± 3 deg
- J1 = 0, J2 = 0.12 m

Joint 0



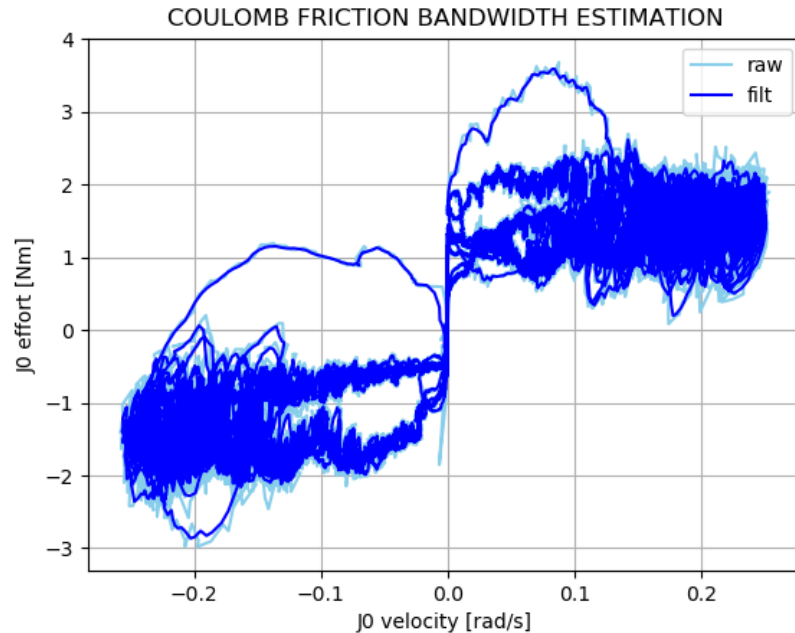
- acc = 0.1 rad/s
- seg_duration = 3s
- center = 0 deg
- amplitude = ± 6.3 deg
- J1 = 0, J2 = 0.12 m

Joint 0



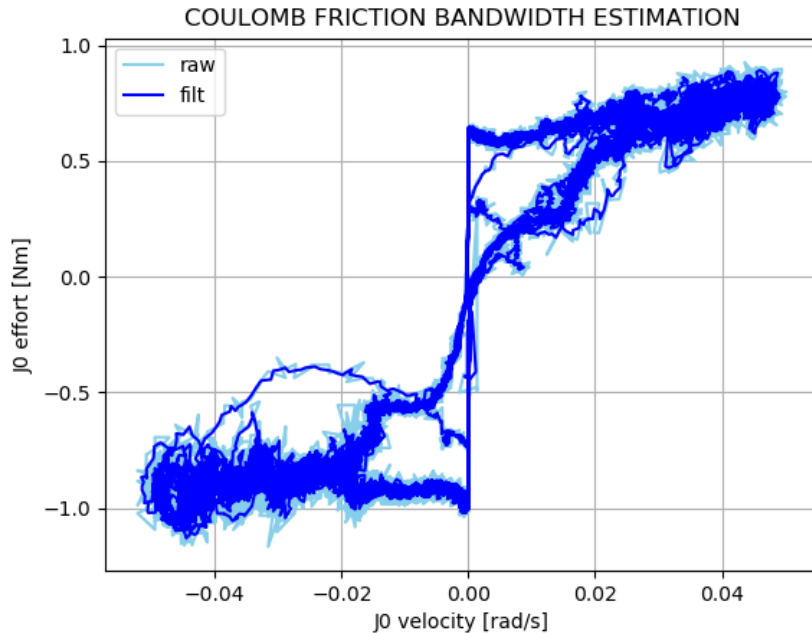
- acc = 0.1 rad/s
- seg_duration = 4s
- center = 0 deg
- amplitude = ± 11.4 deg
- J1 = 0, J2 = 0.12 m

Joint 0



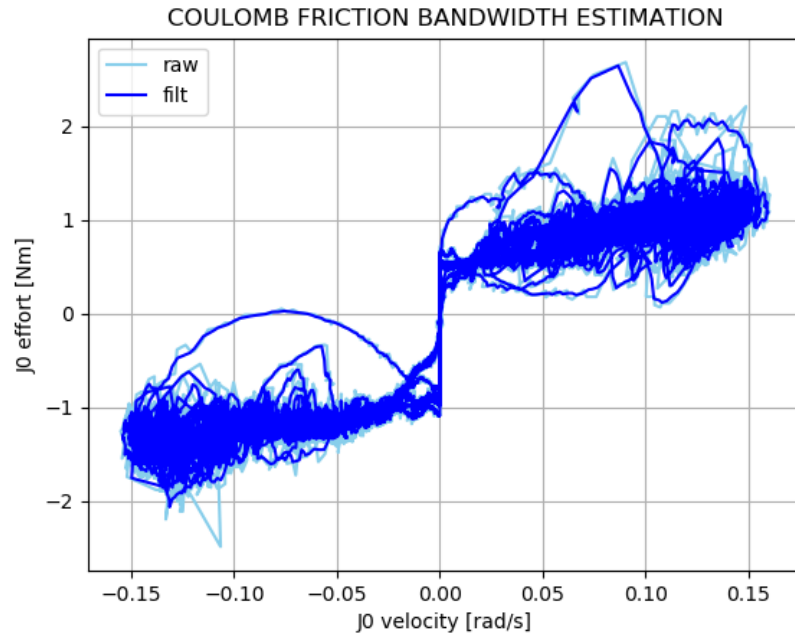
- $\text{acc} = 0.1 \text{ rad/s}$
- $\text{seg_duration} = 5\text{s}$
- $\text{center} = 0 \text{ deg}$
- $\text{Amplitude} = \pm 17.6 \text{ deg}$
- $J1 = 0, J2 = 0.12 \text{ m}$

Joint 0



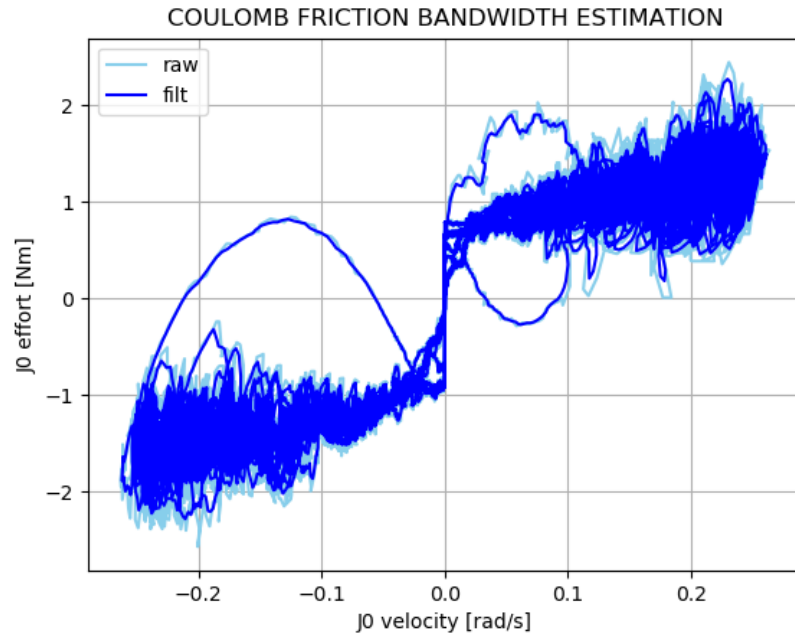
- acc = 0.1 rad/s
- seg_duration = 1s
- center = 0 deg
- Amplitude = ± 0.7 deg
- J1 = -50° , J2 = 0.12 m

Joint 0



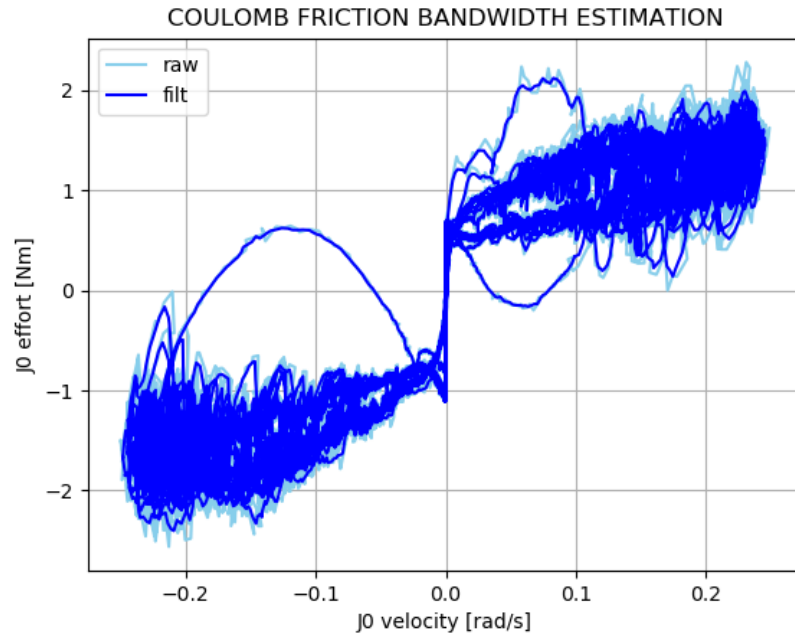
- acc = 0.1 rad/s
- seg_duration = 3s
- center = 0 deg
- Amplitude = ± 6.3 deg
- J1 = -50° , J2 = 0.12 m

Joint 0



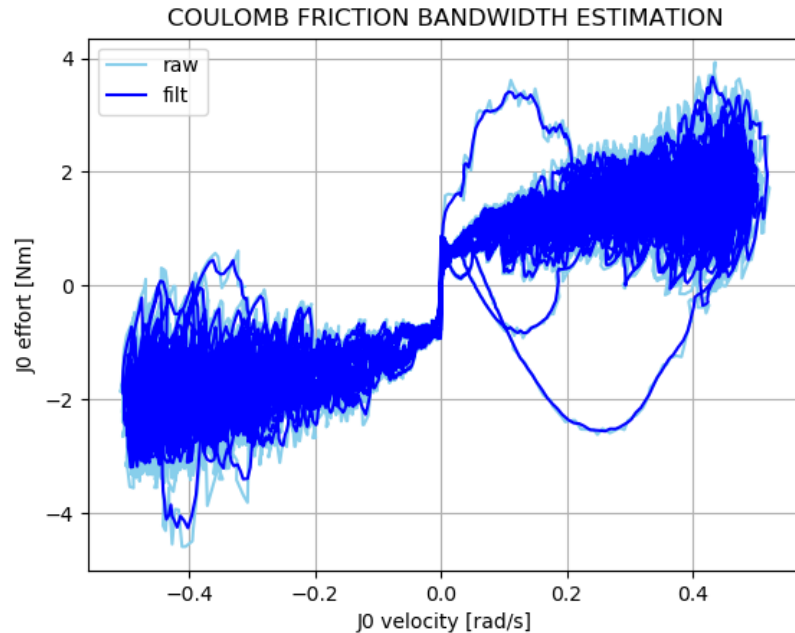
- acc = 0.1 rad/s
- seg_duration = 5s
- center = 0 deg
- Amplitude = ± 17.5 deg
- J1 = -50° , J2 = 0.12 m

Joint 0



- acc = 0.5 rad/s
- seg_duration = 1s
- center = 0 deg
- Amplitude = ± 3.6 deg
- J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s

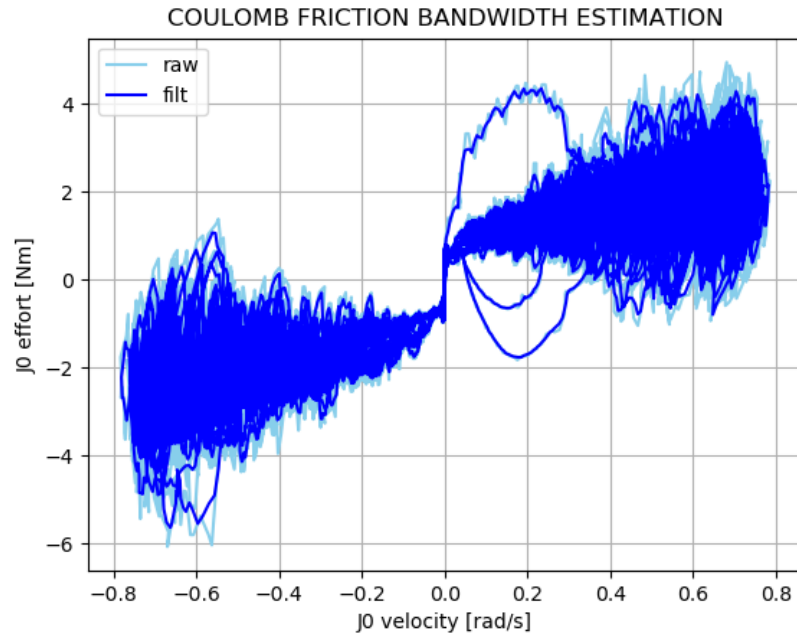
•seg_duration = 2s

•center = 0 deg

•Amplitude = ± 14.2 deg

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

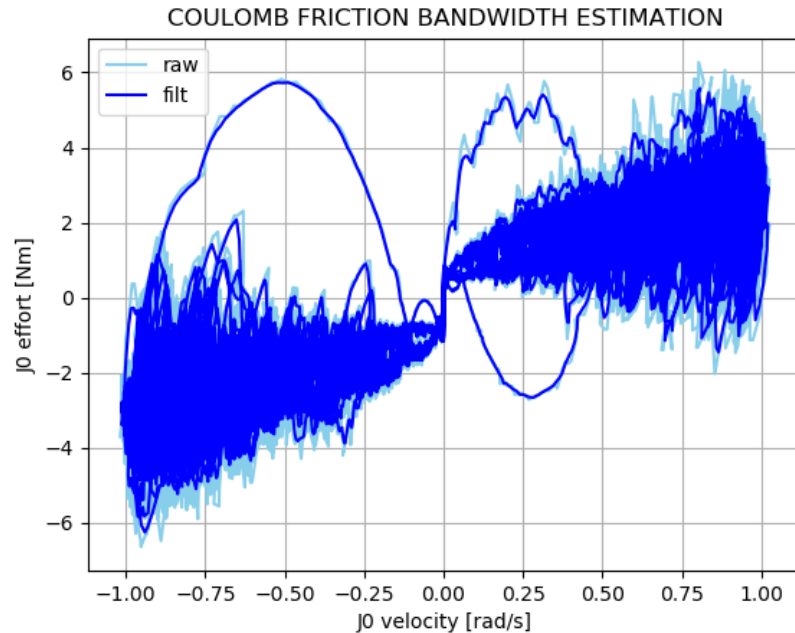
•seg_duration = 3s

•center = 0 deg

•Amplitude = $\pm 31.5 \text{ deg}$

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 1 rad/s²

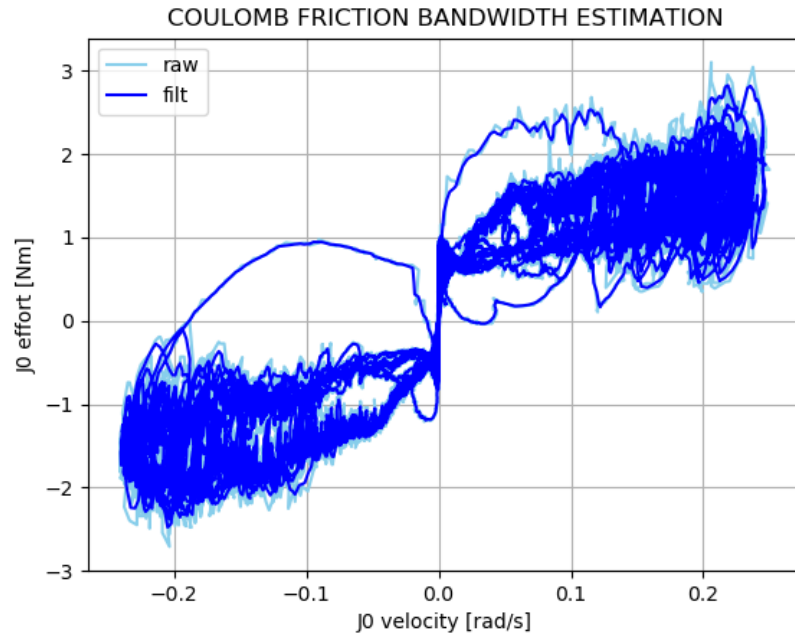
•seg_duration = 2s

•center = 0 deg

•Amplitude = ± 28.5 deg

•J1 = -50°, J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

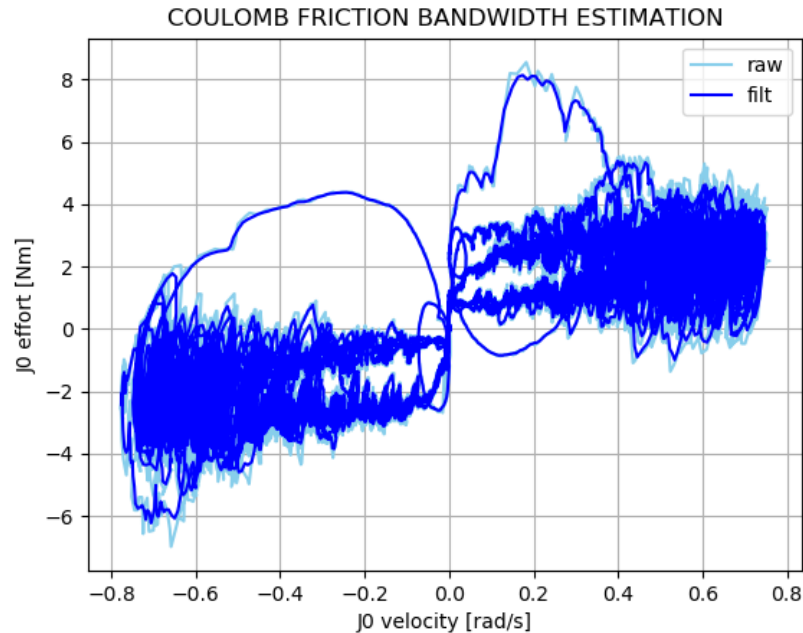
•seg_duration = 1s

•center = 0 deg

•Amplitude = $\pm 3.5 \text{ deg}$

•J1 = 0° , J2 = 0.12 m

Joint 0



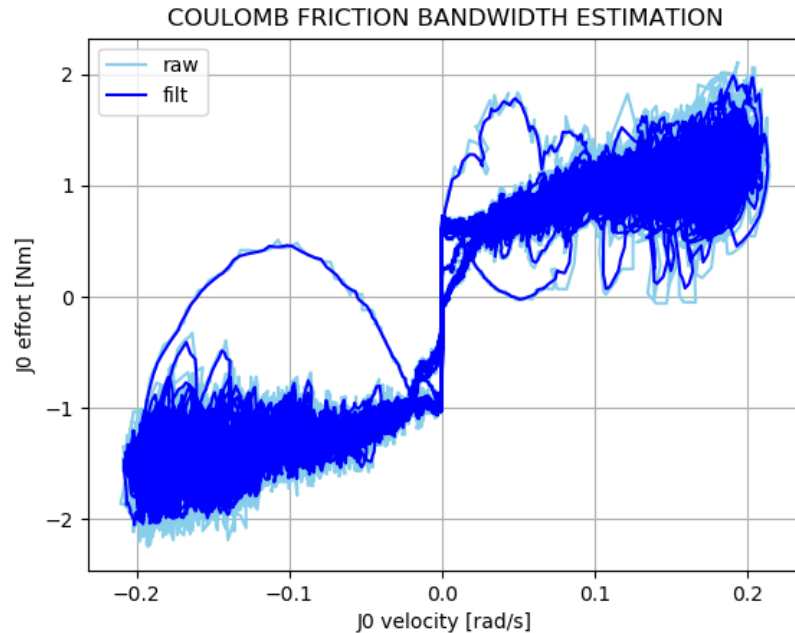
- $\text{acc} = 0.5 \text{ rad/s}^2$
- $\text{seg_duration} = 3\text{s}$
- $\text{center} = 0 \text{ deg}$
- $\text{Amplitude} = \pm 31.5 \text{ deg}$
- $J1 = 0^\circ$, $J2 = 0.12 \text{ m}$

Different central position

- -15 deg (-0.262 rad)
- -30 deg (-0.524 rad)
- -60 deg (-1.047 rad)
- -80 deg (-1.396 rad)
- +20 deg (+0.349 rad)

NOTE: no space for other positive configurations

Joint 0



•acc = 0.1 rad/s^2

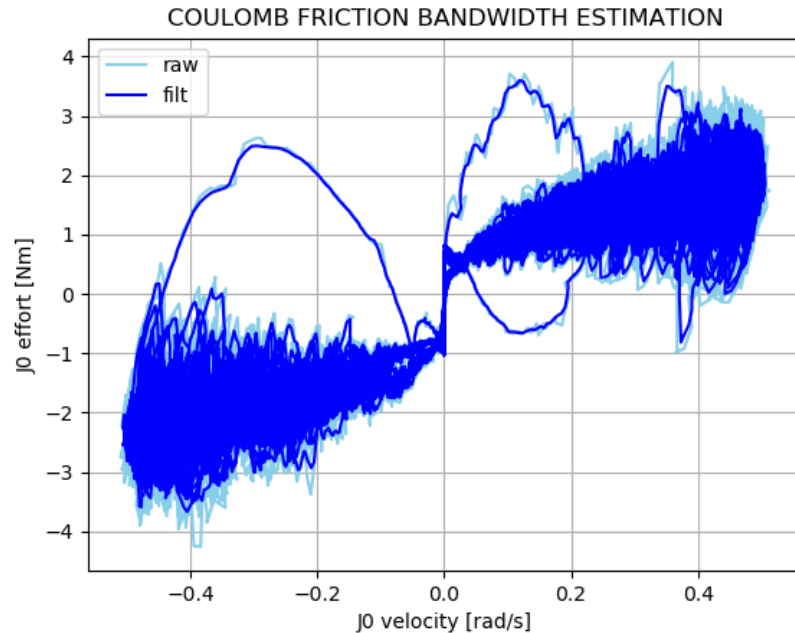
•seg_duration = 4s

•center = -15°

•Amplitude = $\pm 11.5^\circ$

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

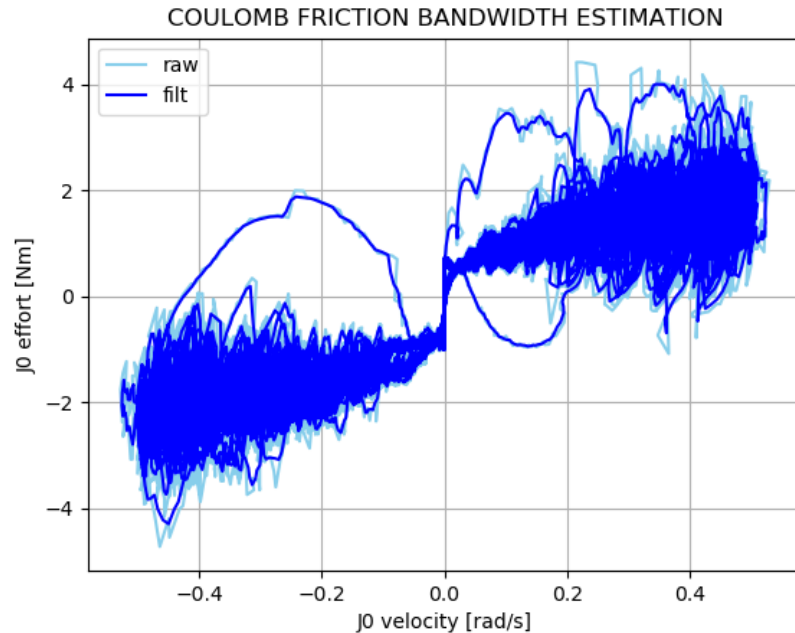
•seg_duration = 2s

•center = -15°

•Amplitude = $\pm 14^\circ$

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

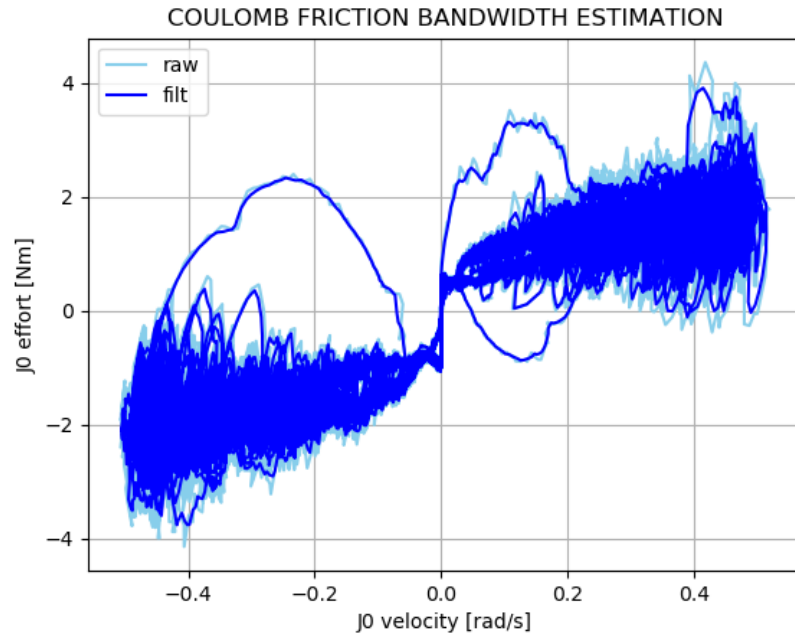
•seg_duration = 2s

•center = -30°

•Amplitude = $\pm 14^\circ$

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

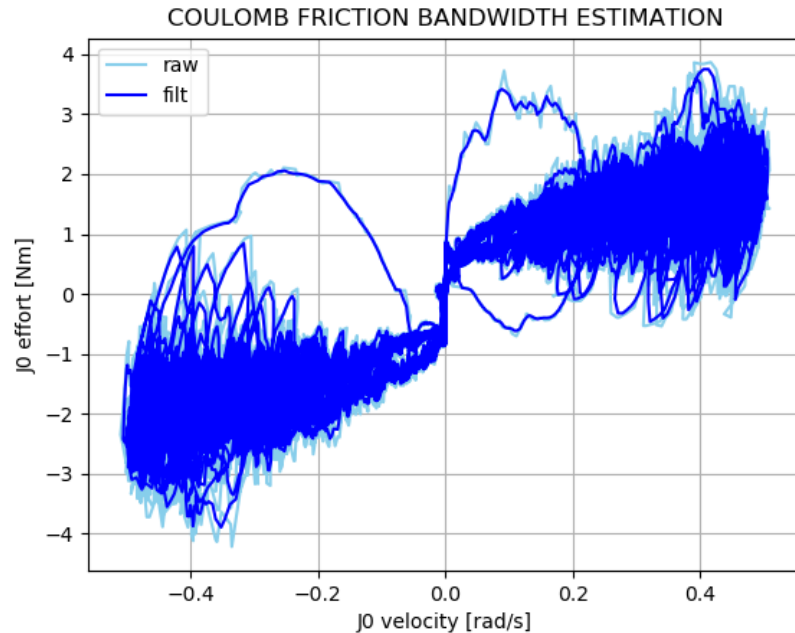
•seg_duration = 2s

•center = -60°

•Amplitude = $\pm 14^\circ$

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

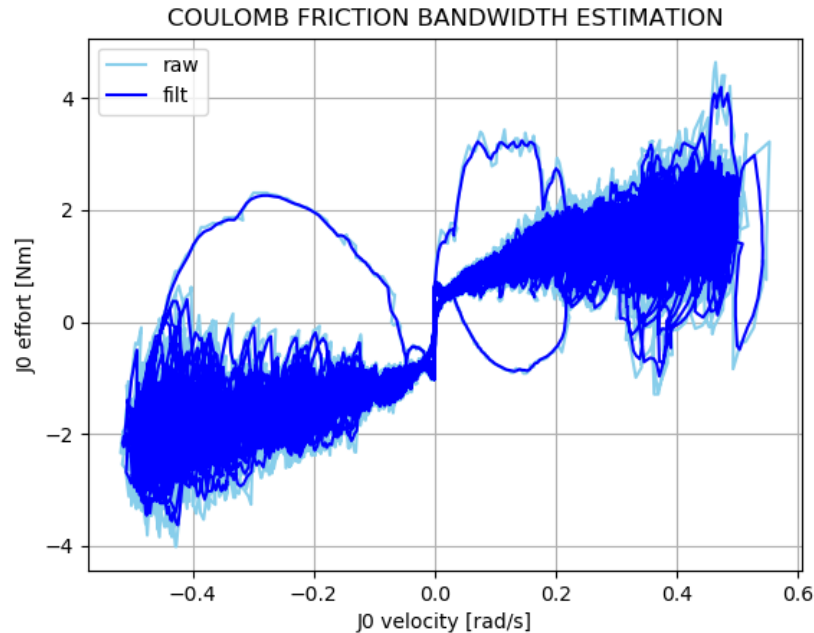
•seg_duration = 2s

•center = -80°

•Amplitude = $\pm 14^\circ$

•J1 = -50° , J2 = 0.12 m

Joint 0



•acc = 0.5 rad/s^2

•seg_duration = 2s

•center = +20 deg

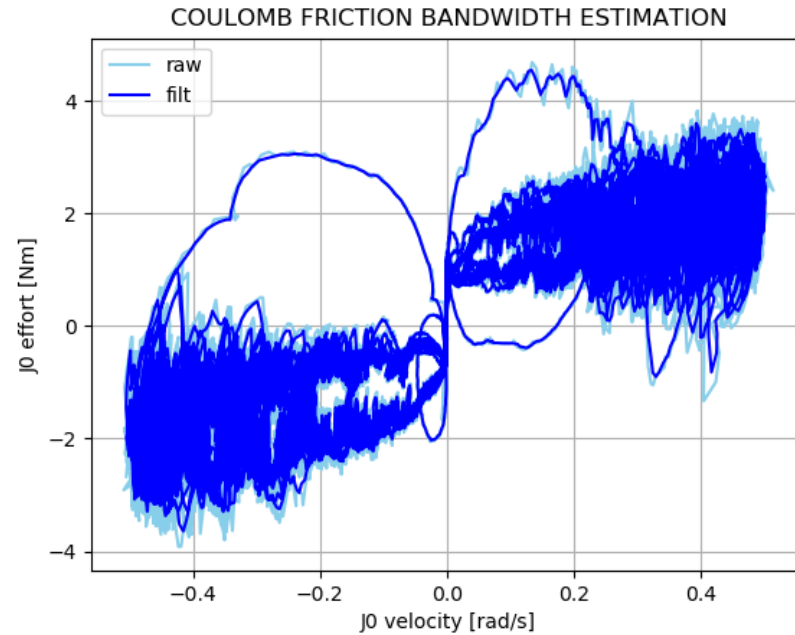
•Amplitude = $\pm 14^\circ$

•J1 = -50° , J2 = 0.12 m

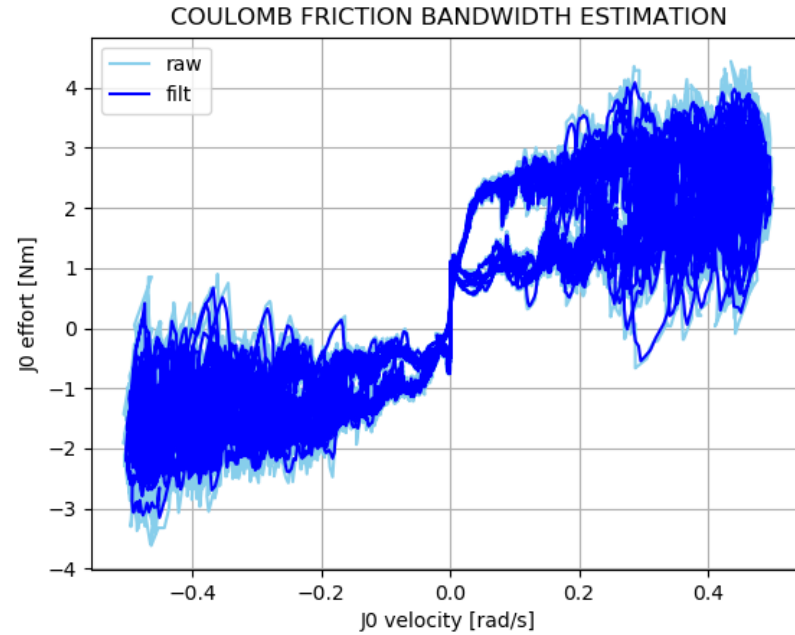
J1 forward different central position

- .J1 = +20 deg
- .Center = 0 deg, -20 deg, -40 deg
- .Acc=0.5 rad/s²
- .Segment duration = 2s → amplitude = ± 14 deg

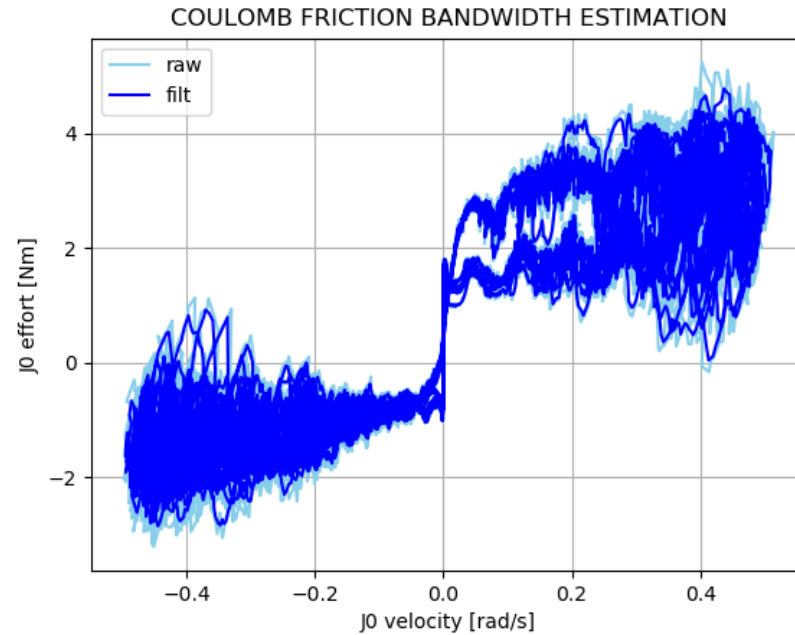
Joint 0 – 0 deg



Joint 0 – -20 deg



Joint 0 – -40 deg



Joint 1

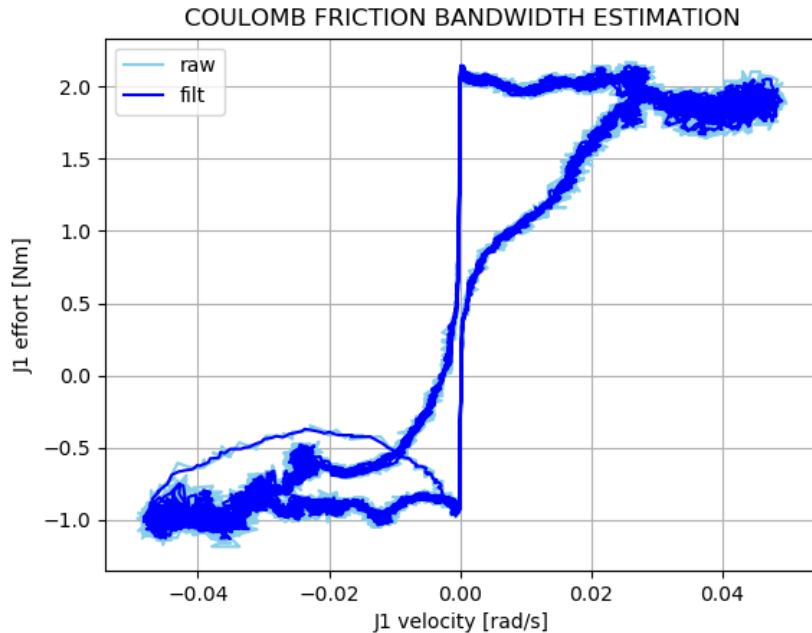
Python frequency = 250 Hz

ROS frequency = 1 Khz

servo_jp

NOTE: if I move J1 to 0 it stays at 0.375 deg when J0=0 (is at 0deg when J0=-90)

Joint 1



•acc = 0.1 rad/s^2

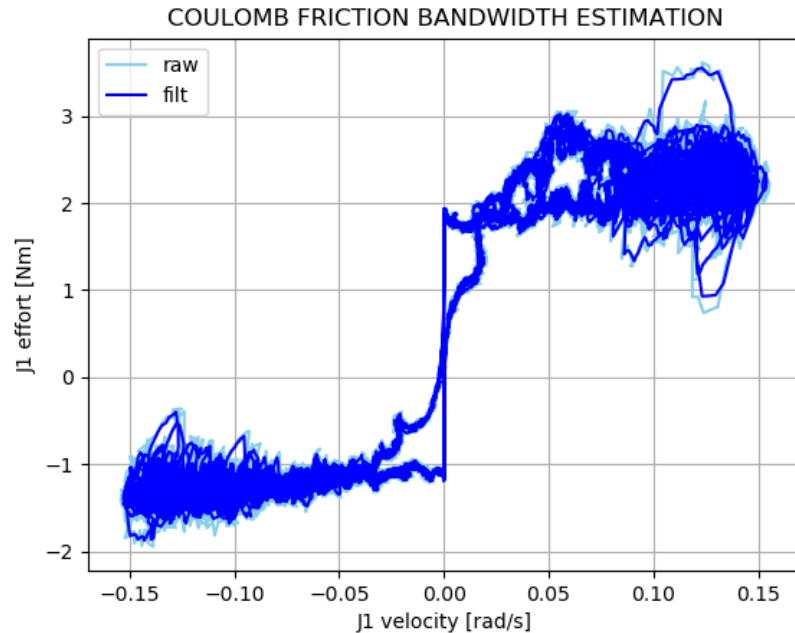
•seg_duration = 1s

•center = 0 deg

•Amplitude = $\pm 0.7^\circ$

•J0 = 0° , J2 = 0.12 m

Joint 1



• $\text{acc} = 0.1 \text{ rad/s}^2$

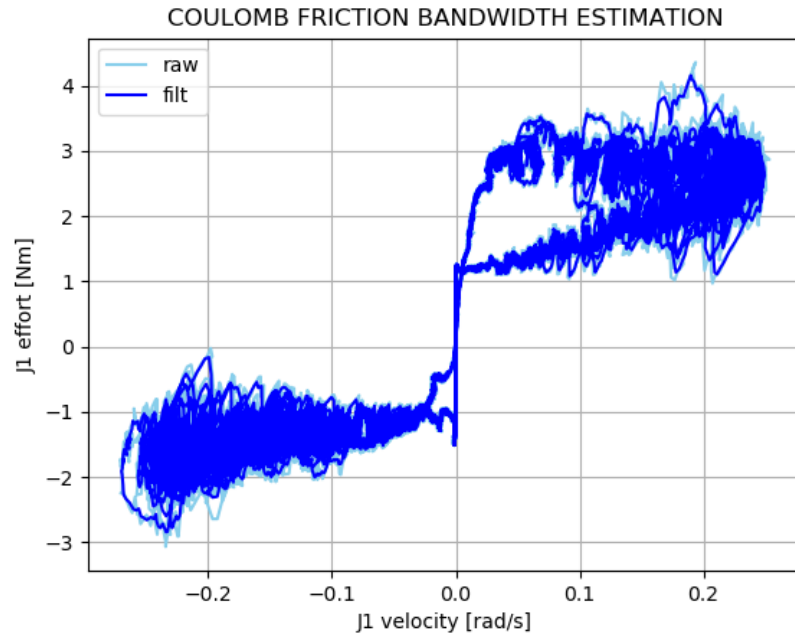
• $\text{seg_duration} = 3\text{s}$

• $\text{center} = 0 \text{ deg}$

• $\text{Amplitude} = \pm 6.3^\circ$

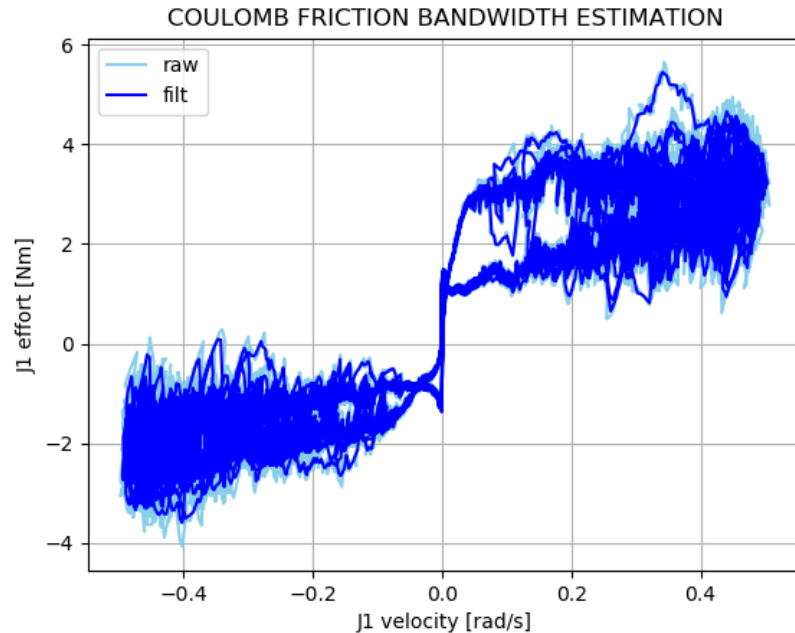
• $J0 = 0^\circ$, $J2 = 0.12 \text{ m}$

Joint 1



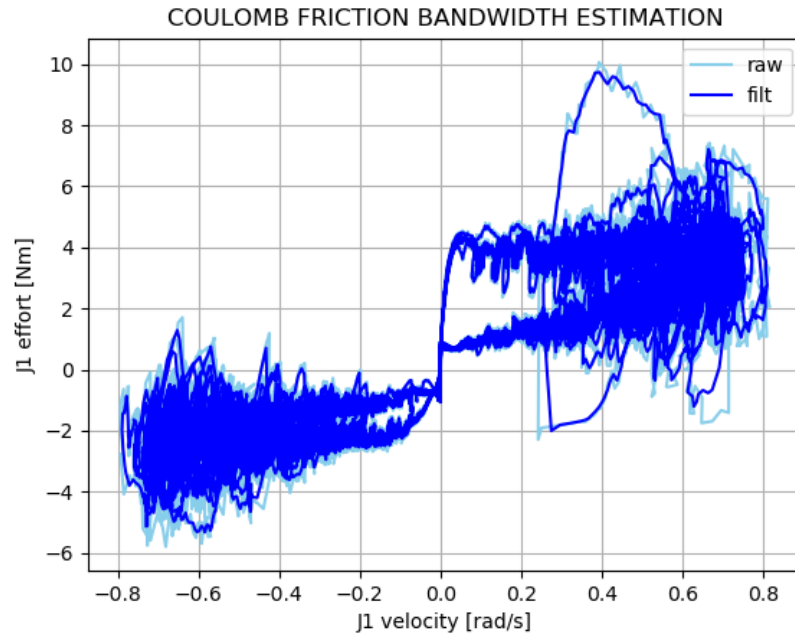
- $\text{acc} = 0.1 \text{ rad/s}^2$
- $\text{seg_duration} = 5\text{s}$
- $\text{center} = 0 \text{ deg}$
- $\text{Amplitude} = \pm 17.6^\circ$
- $J0 = 0^\circ$, $J2 = 0.12 \text{ m}$

Joint 1



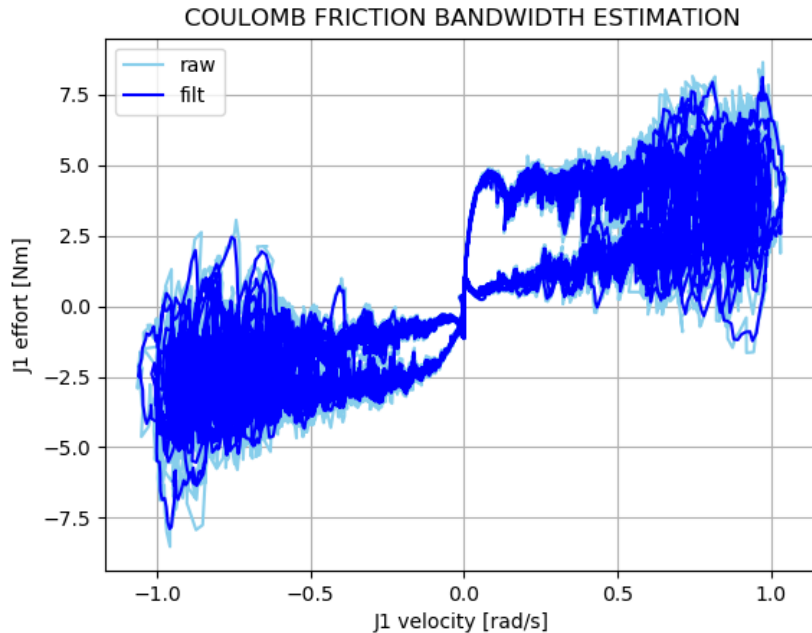
- $\text{acc} = 0.5 \text{ rad/s}^2$
- $\text{seg_duration} = 2\text{s}$
- $\text{center} = 0 \text{ deg}$
- $\text{Amplitude} = \pm 14.1^\circ$
- $J0 = 0^\circ$, $J2 = 0.12 \text{ m}$

Joint 1



- $\text{acc} = 0.5 \text{ rad/s}^2$
- $\text{seg_duration} = 4\text{s}$
- $\text{center} = 0 \text{ deg}$
- $\text{Amplitude} = \pm 31.7^\circ$
- $J0 = 0^\circ$, $J2 = 0.12 \text{ m}$

Joint 1



•acc = 1 rad/s²

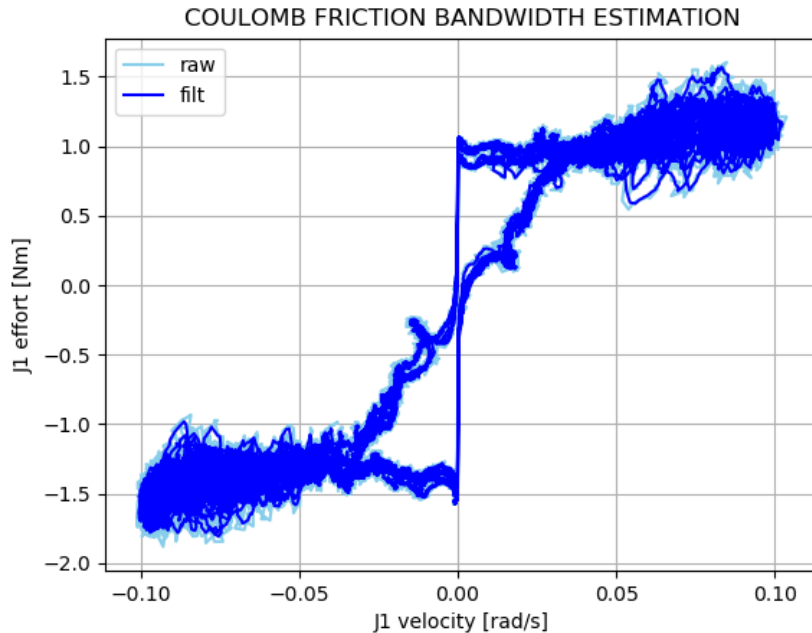
•seg_duration = 2s

•center = 0 deg

•Amplitude = $\pm 28.2^\circ$

•J0 = 0°, J2 = 0.12 m

Joint 1



•acc = 0.1 rad/s^2

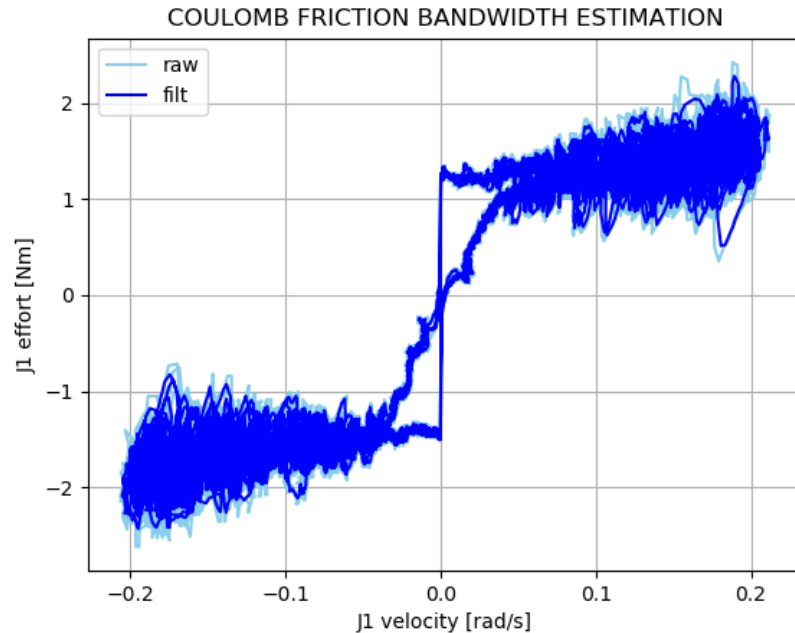
•seg_duration = 2s

•center = 0 deg

•Amplitude = $\pm 2.9^\circ$

•J0 = -90° , J2 = 0.12 m

Joint 1



•acc = 0.1 rad/s^2

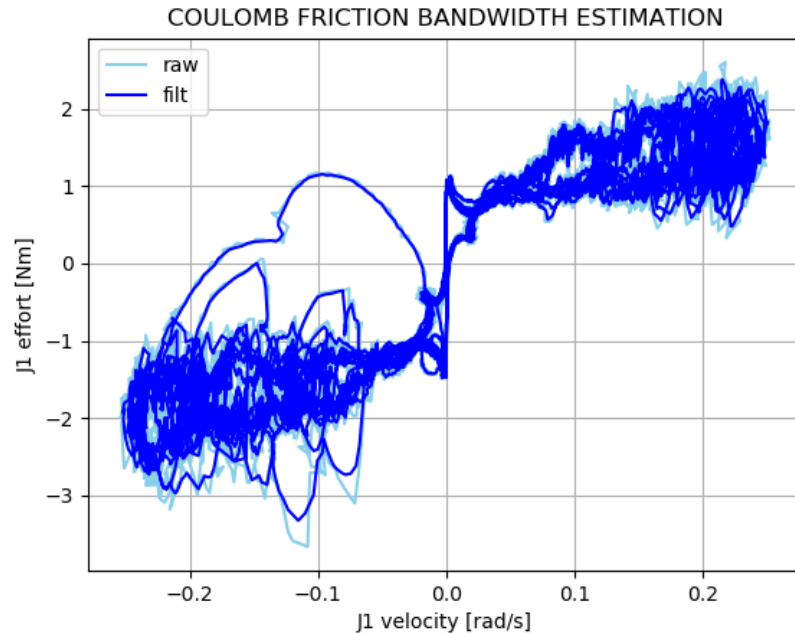
•seg_duration = 4s

•center = 0 deg

•Amplitude = $\pm 11.5^\circ$

•J0 = -90° , J2 = 0.12 m

Joint 1



•acc = 0.5 rad/s^2

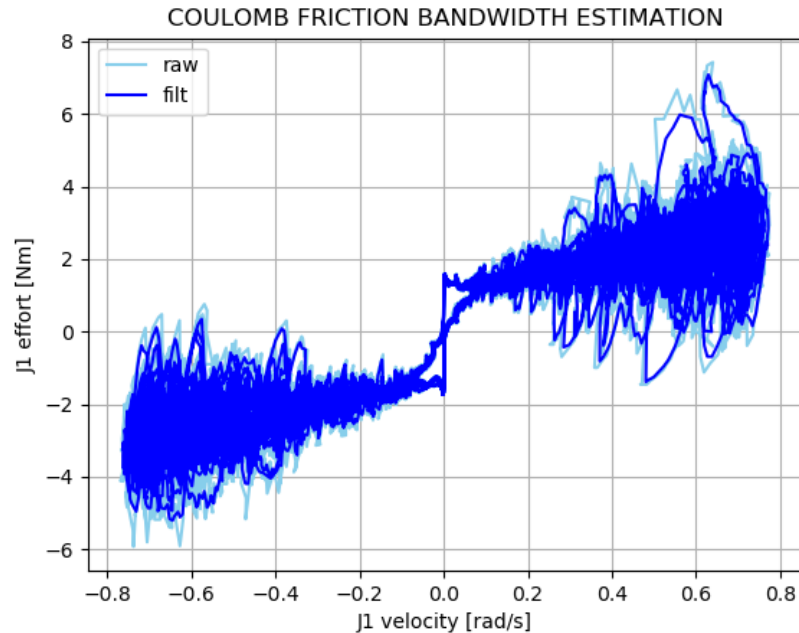
•seg_duration = 1s

•center = 0 deg

•Amplitude = $\pm 3.6^\circ$

•J0 = -90° , J2 = 0.12 m

Joint 1



•acc = 0.5 rad/s^2

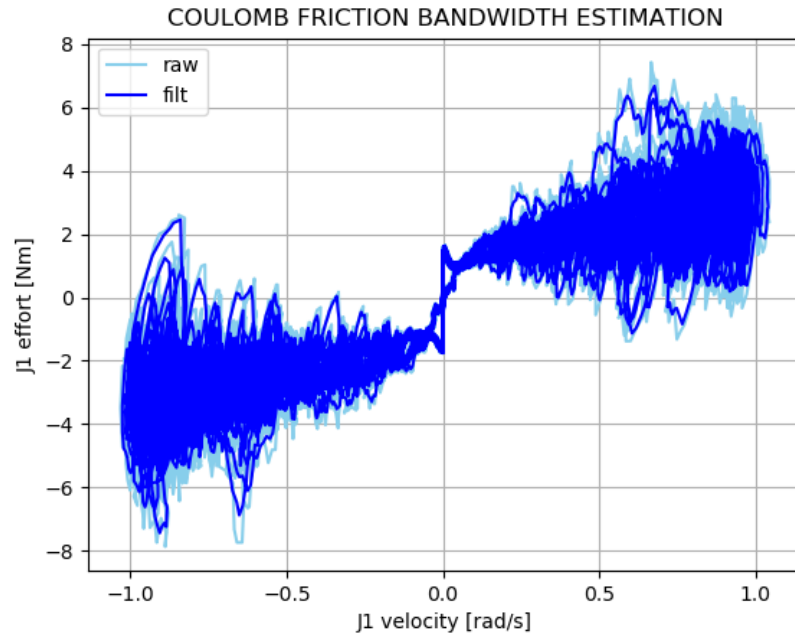
•seg_duration = 3s

•center = 0 deg

•Amplitude = $\pm 32.2^\circ$

•J0 = -90° , J2 = 0.12 m

Joint 1



•acc = 1 rad/s²

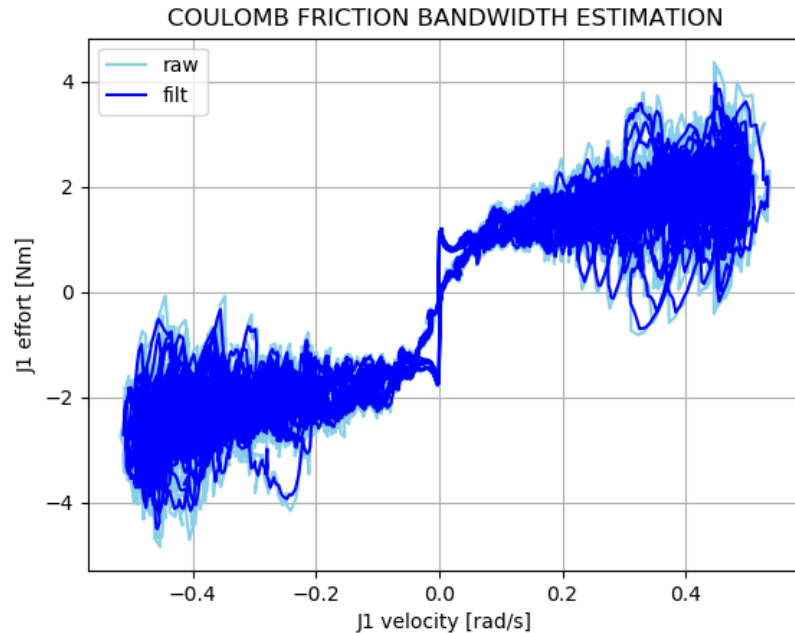
•seg_duration = 2s

•center = 0 deg

•Amplitude = $\pm 28.5^\circ$

•J0 = -90°, J2 = 0.12 m

Joint 1



• $\text{Acc} = 0.5 \text{ rad/s}^2$

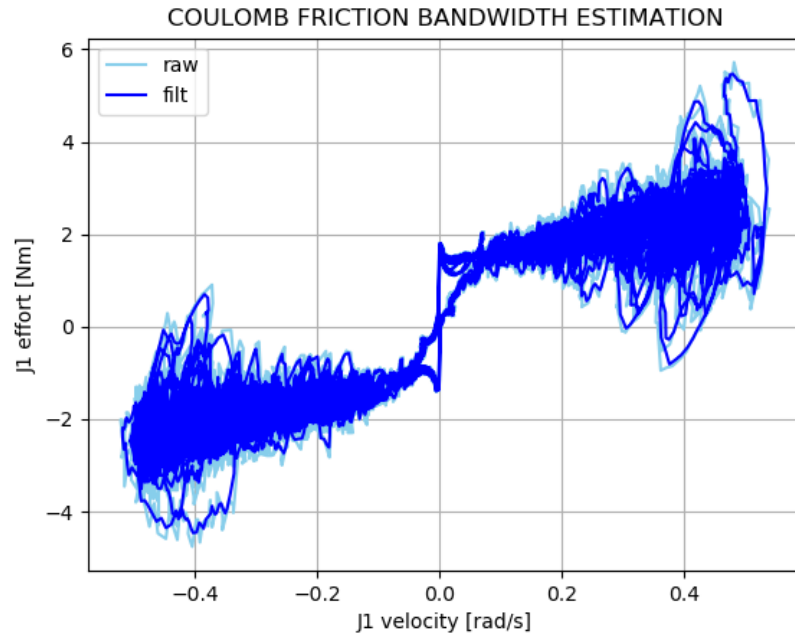
• $\text{seg_duration} = 2\text{s}$

• $\text{center} = -20 \text{ deg}$

• $\text{Amplitude} = \pm 14^\circ$

• $J0 = -90^\circ$, $J2 = 0.12 \text{ m}$

Joint 1



• $\text{Acc} = 0.5 \text{ rad/s}^2$

• $\text{seg_duration} = 2\text{s}$

• $\text{center} = 20 \text{ deg}$

• $\text{Amplitude} = \pm 14^\circ$

• $J0 = -90^\circ$, $J2 = 0.12 \text{ m}$

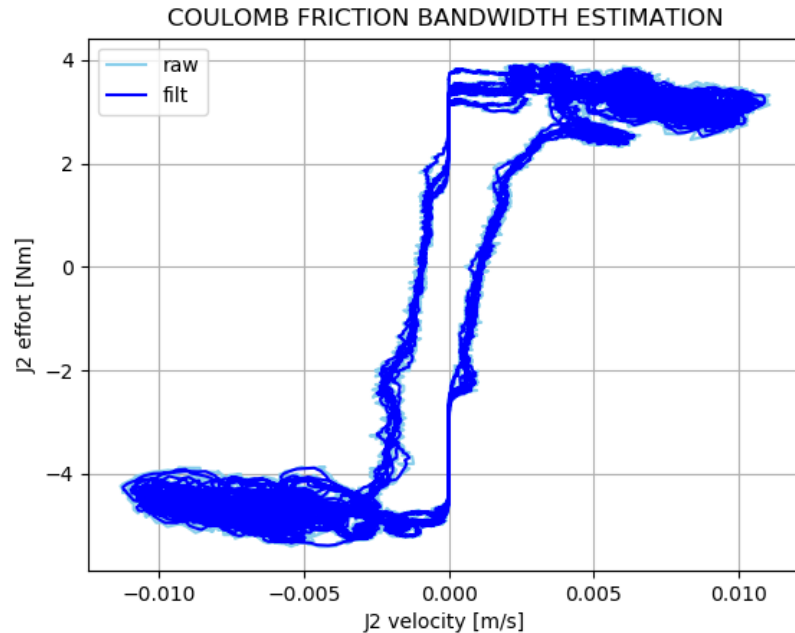
Joint 3

.Python frequency = 250 Hz

.ROS frequency = 1 Khz

.servo_jp

Joint 3



• $\text{Acc} = 0.01 \text{ m/s}^2$

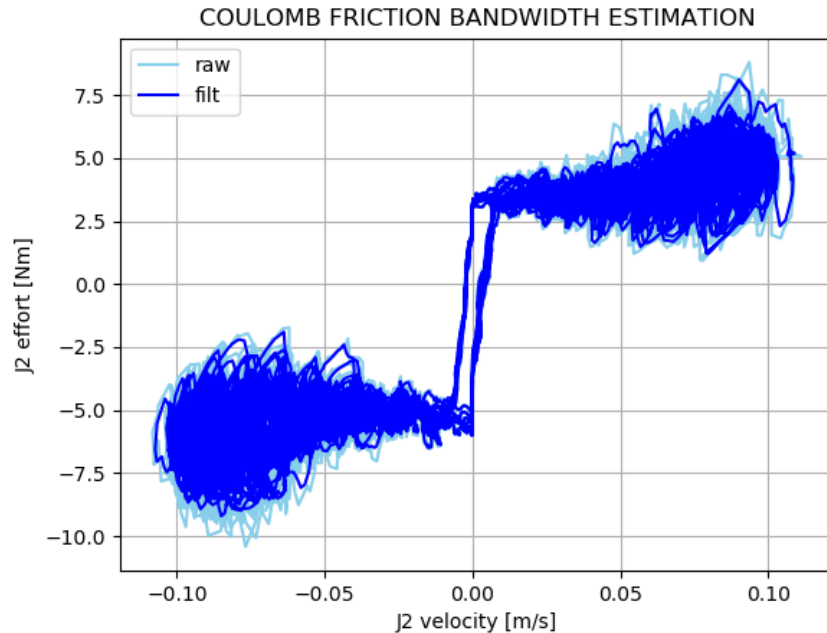
• $\text{seg_duration} = 2\text{s}$

• $\text{Center} = 0.118 \text{ m}$

• $\text{Amplitude} = \pm 0.005 \text{ m}$

• $J0 = 0^\circ, J1 = 0^\circ$

Joint 3



• $\text{Acc} = 0.01 \text{ m/s}^2$

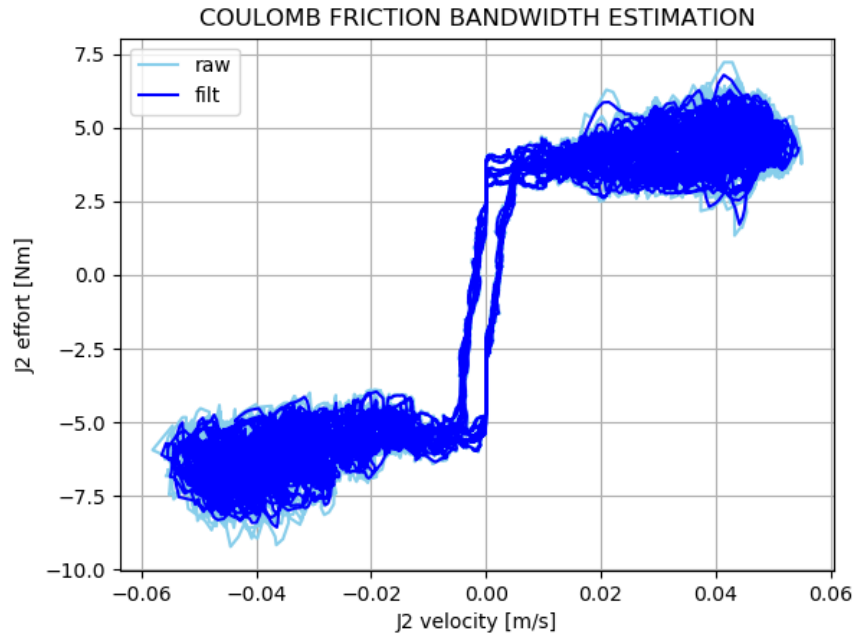
• $\text{seg_duration} = 4\text{s}$

• $\text{Center} = 0.090 \text{ m}$

• $\text{Amplitude} = \pm 0.020 \text{ m}$

• $J0 = 0^\circ$, $J1 = 0^\circ$

Joint 3



• $\text{Acc} = 0.05 \text{ m/s}^2$

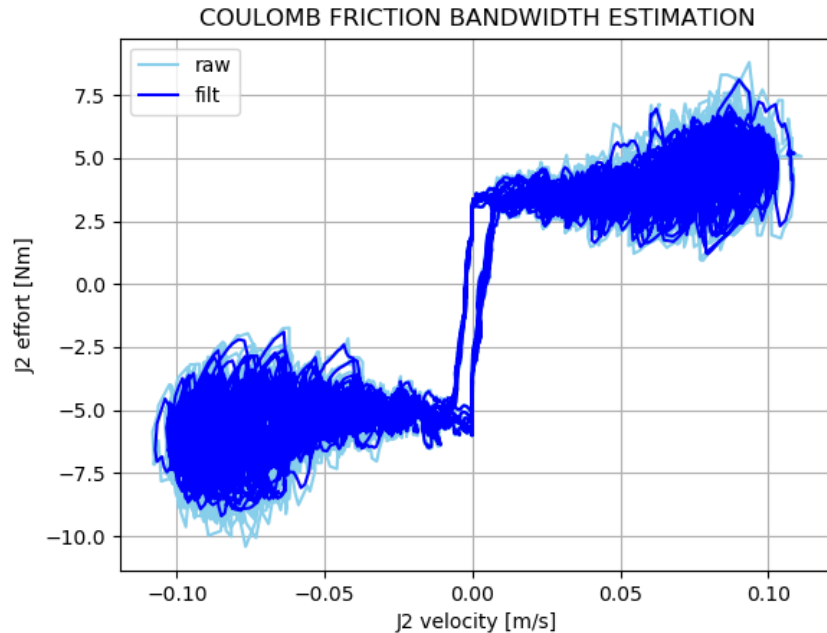
• $\text{seg_duration} = 2\text{s}$

• $\text{Center} = 0.170 \text{ m}$

• $\text{Amplitude} = \pm 0.025 \text{ m}$

• $J0 = 0^\circ$, $J1 = 0^\circ$

Joint 3



- $\text{Acc} = 0.1 \text{ m/s}^2$
- $\text{seg_duration} = 2\text{s}$
- $\text{Center} = 0.150 \text{ m}$
- $\text{Amplitude} = \pm 0.050 \text{ m}$
- $J0 = 0^\circ, J1 = 0^\circ$